Spatial diagnostics and network workflows

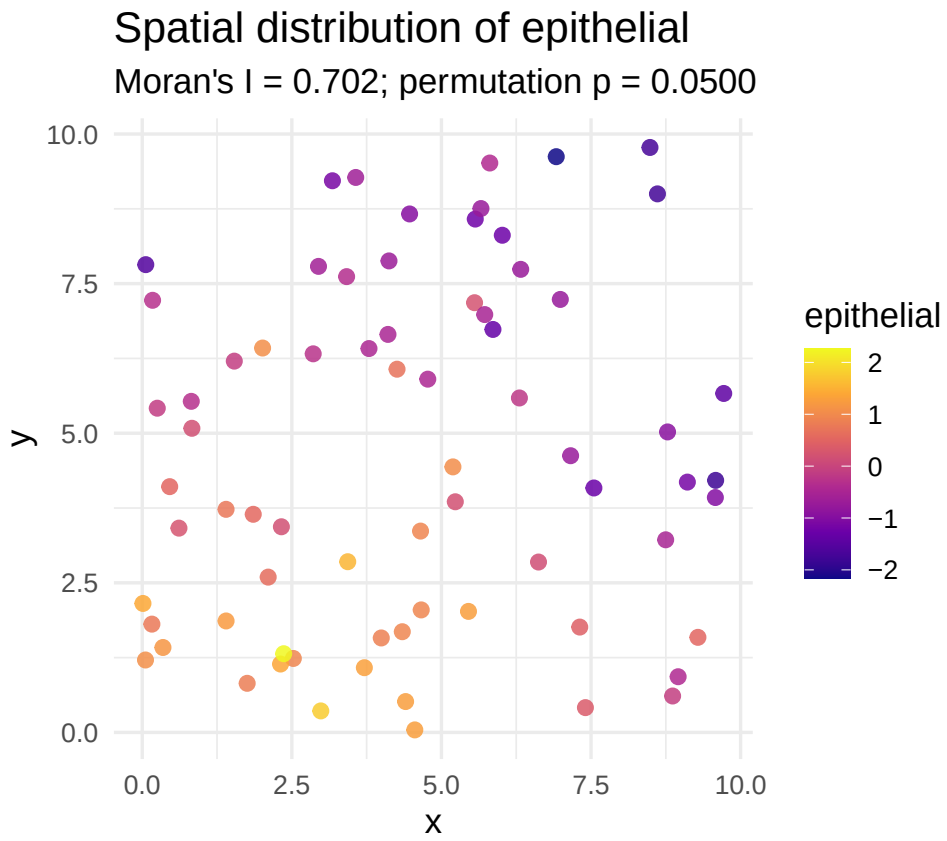
This vignette exercises the spatial and network analysis family with 72 synthetic observations. The small permutation count and compact data are intended for reproducibility checks, not final analysis settings.

```
d <- demo_spatial_data(n = 72, seed = 2026)
features <- c("epithelial", "APC", "CTL", "Treg", "T_helper")
mapping <- list(
  x = "x", y = "y", z = NULL, zone = "zone", id = "sample_id",
  outcome = "disease", exposure = "epithelial", mediator = "APC"
)
```

Spatial exploration and autocorrelation

```
spatial_qc <- run_analysis(
  "spatial_qc", d, mapping, features,
  list(feature = "epithelial", neighbors = 6, permutations = 19, seed = 2026)
)
spatial_qc$table
#>                  metric      value
#> 1 Complete observations 72.0000000
#> 2             Neighbors  6.0000000
#> 3             Moran's I  0.7019779
#> 4   Permutation p-value  0.0500000
```

```
spatial_qc$plot
```

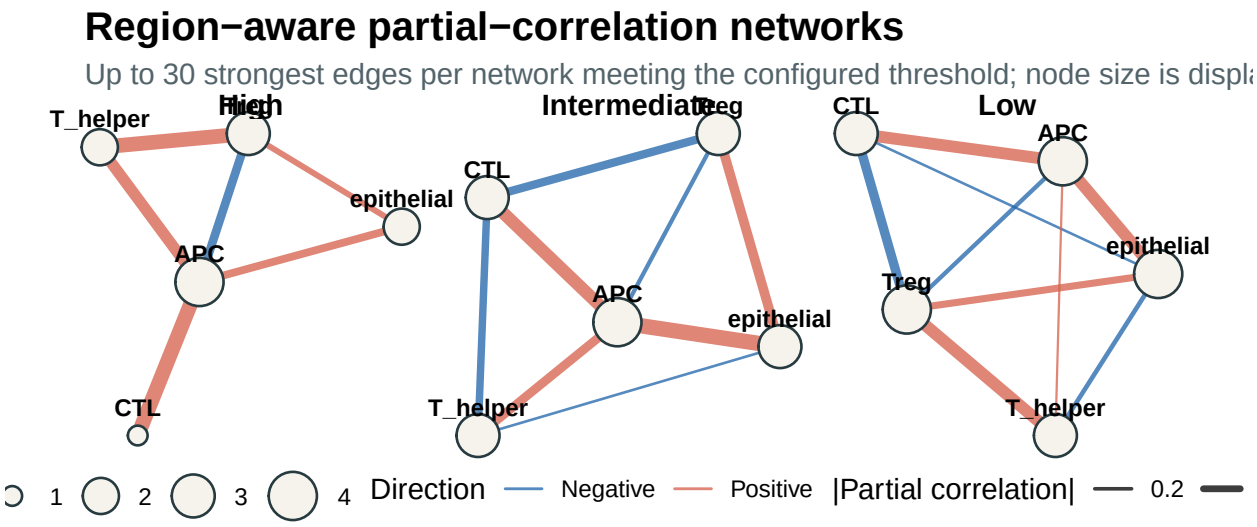


The output reports the configured neighbor graph, statistic, permutation calculation, and provenance. It does not establish a biological mechanism.

Region-aware conditional network

```
conditional <- run_analysis(
  "conditional_network", d, mapping, features,
  list(threshold = 0.15, ridge = 0.05, by_zone = TRUE)
)
head(conditional$table, 10)
#>        network     source   target     weight     sign
#> High.1    High epithelial      APC  0.3318986 Positive
#> High.2    High        APC      CTL  0.5554517 Positive
#> High.3    High epithelial     Treg  0.2728543 Positive
#> High.4    High        APC     Treg -0.3581961 Negative
#> High.5    High        APC T_helper  0.4468269 Positive
#> High.6    High       Treg T_helper  0.5542829 Positive
#> Low.1      Low epithelial      APC  0.4614824 Positive
#> Low.2      Low epithelial      CTL -0.1663843 Negative
#> Low.3      Low        APC      CTL  0.4633987 Positive
#> Low.4      Low epithelial     Treg  0.3099184 Positive
conditional$summary
#>      networks visible_edges     threshold         ridge
#>          3.00         23.00          0.15          0.05
```

```
conditional$plot
```



The stored matrices can be audited directly.

```
names(conditional$matrices)
#> [1] "High"         "Low"          "Intermediate"
vapply(conditional$matrices, nrow, integer(1))
#>         High          Low Intermediate
#>            5            5            5
```

Edges are regularized conditional-association summaries. The vignette intentionally stops at reproducible output inspection.

Public advanced engines

The following calls use the same 72-row table. They are not evaluated while building the vignette because runtime depends on the installed public engines and local hardware.

ISPat 2D

The ISPat 2D path calls `ISPAT3D::ispat3d_fit_2d()` from the CRAN package ISPAT3D (0.3.1 or newer): a Matérn-3/2 Vecchia GP is fitted per marker within each zone-section group, and a shared-plus-zone factor covariance is fitted by Gaussian maximum likelihood. The small anchor and iteration settings below are software smoke values; the package defaults (rank 5, up to 5,000 anchors, 15 neighbors, 30 GP iterations) are the analysis settings. Because `sample_id` is unique per row it is not treated as a section label, and the result notes say so.

```r
install.packages("ISPAT3D")

ispat <- run_analysis(
  "ispat", d, mapping, features,
  list(
    rank = 2L, anchor_min = 12L, anchor_max = 24L, gp_neighbors = 5L,
    gp_maxit = 3L, factor_maxit = 60L, threads = 1L, seed = 2026L,
    threshold = 0.05
  )
)
ispat$summary
names(ispat$matrices)      # one signed partial-correlation matrix per zone
head(ispat$table)
table(ispat$gp_log$status)
ispat$notes
ispat$plot
```

GP-GHS

The short chain below is a software smoke check. Increase Monte Carlo settings and perform convergence assessment for research use. GPGHS 1.0 selects a predictor when its group shrinkage factor falls below `edge_threshold` (0.5 in the article) and combines the two directed decisions with the AND rule.

```r
remotes::install_github("sagnikbhadury/GP-GHS")

gpghs <- run_analysis(
  "gpghs", d, mapping, features,
  list(
    basis = 3L, nu = 1.5, nmc = 200L, burn = 50L, thin = 2L,
    edge_threshold = 0.5, symmetry = "AND", min_detect = 0, cores = 1L, seed = 1L
  )
)
gpghs$summary
head(gpghs$table)          # GPGHS score, sign, sign change, and RMS weight per selected edge
gpghs$plot
```

The GP-GHS graph uses the fitted symmetric adjacency for topology and the smaller directional RMS of the posterior mean coefficient fields for edge width. It does not use kappa as a visual weight, and it does not assign a single sign to a field that can change sign across space; the edge table carries the GPGHS sign label instead.

Use `method_citations("ispat")` and `method_citations("gpghs")` to retrieve the required software and method references.